

Q&A Dataset: Mambakkam Resonant Cave System Report

This dataset contains complex questions based *only* on the "Preliminary Report: Mambakkam Resonant Cave System (MRCS) - Phase 1 Survey". Use it to evaluate your RAG system's accuracy against the Ground Truth. Efficiency can be assessed by measuring query latency during testing.

Question	Ground Truth Answer (derived only from the report text)	Complexity Aspects Tested
Explain the conditions under which Sonite crystal resonance intensity would be significantly reduced, citing the specific location and causal agent mentioned.	Sonite resonance intensity is significantly dampened in Chamber C2 when the concentration of Argonite gas (ARG) exceeds 22 ppm, as ARG is suggested to act as an acoustic inhibitor for Sonite.	Causality, Specificity, Location Dependency
Describe the two distinct factors that influence the observed color of Hydro-Luminite veins and state where this interaction complicates measurements.	The color shifts based on water temperature (blue <10°C, green >13°C). This complicates direct intensity-purity readings in Chamber C3 because the water temperature there is stable at 14°C (causing a green color).	Identifying Multiple Factors, Understanding Interaction, Location Specificity
What is the proposed mechanism (Hypothesis H3) explaining why Sonite resonance affects Hydro-Luminite brightness, and what type of testing is needed to verify it?	H3 proposes it's a secondary piezoelectric effect within the Hydro-Luminite itself, triggered by specific infrasound frequencies from Sonite. Verification requires controlled frequency exposure tests on Hydro-Luminite samples.	Understanding Hypotheses, Identifying Required Validation Methods
Under what specific, measurable condition does the concentration of Ferro-Resonant Dust in the C3 aquifer negatively impact Hydro-Luminite brightness, and why?	When the aquifer flow rate exceeds 0.5 m/s (measured by MISS-WAT-F01), the dust concentration in suspension increases dramatically. This negatively impacts Hydro-Luminite brightness	Multi-step Causality, Threshold Condition, Instrument Reference, Linking Mineral Properties

	because the dust has high iron content, and increased dissolved iron reduces brightness.	
Contrast the atmospheric challenges requiring specific safety protocols in Chamber C2 versus Chamber C3.	C2 requires monitoring of Argonite gas (ARG), with the sensor needing frequent recalibration. C3 requires supplementary breathing apparatus due to significantly lower O2 levels (17.5% avg.) and high humidity.	Comparison, Identifying Distinct Conditions, Safety Protocol Linkage
If the MISS-OPT-P02 photometer readings are saturated in C2, what two simultaneous conditions are most likely occurring according to the report?	Saturation likely occurs during peak Sonite-boosted Hydro-Luminite events. This implies: 1) Sonite infrasound resonance is above 75 dB, and 2) Hydro-Luminite is present (likely near water seepages).	Inference, Combining Conditions, Instrument Limitation Context
According to the report, how might external weather phenomena indirectly influence the acoustic properties within Chamber C2?	External barometric pressure inversely correlates with ARG concentration in C2. Since ARG concentration above 22 ppm dampens Sonite resonance, changes in external barometric pressure could indirectly modulate Sonite resonance intensity in C2 by altering ARG levels.	Multi-hop Reasoning (Pressure -> ARG -> Sonite Resonance), Indirect Causality
What unresolved question exists regarding the physical state and composition of Ferro-Resonant Dust within the cave system?	UQ2 asks whether the composition of the dust differs significantly between its airborne state in C1 and its water-suspended state in C3.	Identifying Unresolved Questions, State Dependency
Based on Hypothesis H1 and the introduction, what sequence of events is proposed to explain the presence of Argonite gas in Chamber C2?	The Feb 12 seismic event (TN-SEIS-2025-0212) exposed a fissure providing access. H1 proposes ARG outgassing originates from deep strata connected via micro-fissures, potentially linked to this seismic event	Sequencing Events, Linking Hypothesis to Introduction

	trigger path.	
Explain how the accuracy of water purity measurements using Hydro-Luminate brightness in C3 could be compromised by both aquifer dynamics and instrument limitations.	Aquifer dynamics: Flow rate > 0.5 m/s increases Ferro-Resonant Dust suspension, raising dissolved iron and reducing brightness independent of actual purity. Instrument limitations: The MISS-WAT-F01 flowmeter used to measure this trigger condition loses accuracy when dust suspension exceeds 50 NTU.	Synthesis of Multiple Factors, Linking Dynamics to Measurement, Instrument Limitations

CSV Format for Evaluation:

Question,Ground Truth Answer

"Explain the conditions under which Sonite crystal resonance intensity would be significantly reduced, citing the specific location and causal agent mentioned."; "Sonite resonance intensity is significantly dampened in Chamber C2 when the concentration of Argonite gas (ARG) exceeds 22 ppm, as ARG is suggested to act as an acoustic inhibitor for Sonite."

"Describe the two distinct factors that influence the observed color of Hydro-Luminate veins and state where this interaction complicates measurements."; "The color shifts based on water temperature (blue <10°C, green >13°C). This complicates direct intensity-purity readings in Chamber C3 because the water temperature there is stable at 14°C (causing a green color)."

"What is the proposed mechanism (Hypothesis H3) explaining why Sonite resonance affects Hydro-Luminate brightness, and what type of testing is needed to verify it?"; "H3 proposes it's a secondary piezoelectric effect within the Hydro-Luminate itself, triggered by specific infrasound frequencies from Sonite. Verification requires controlled frequency exposure tests on Hydro-Luminate samples."

"Under what specific, measurable condition does the concentration of Ferro-Resonant Dust in the C3 aquifer negatively impact Hydro-Luminate brightness, and why?"; "When the aquifer flow rate exceeds 0.5 m/s (measured by MISS-WAT-F01), the dust concentration in suspension increases dramatically. This negatively impacts Hydro-Luminate brightness because the dust has high iron content, and increased dissolved iron reduces brightness."

"Contrast the atmospheric challenges requiring specific safety protocols in Chamber C2 versus Chamber C3."; "C2 requires monitoring of Argonite gas (ARG), with the

sensor needing frequent recalibration. C3 requires supplementary breathing apparatus due to significantly lower O2 levels (17.5% avg.) and high humidity."

"If the MISS-OPT-PO2 photometer readings are saturated in C2, what two simultaneous conditions are most likely occurring according to the report?"; "Saturation likely occurs during peak Sonite-boosted Hydro-Luminite events. This implies: 1) Sonite infrasound resonance is above 75 dB, and 2) Hydro-Luminite is present (likely near water seepages)."

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"What unresolved question exists regarding the physical state and composition of Ferro-Resonant Dust within the cave system?"; "UQ2 asks whether the composition of the dust differs significantly between its airborne state in C1 and its water-suspended state in C3."

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This dataset should provide a rigorous testbed. The complexity lies in the multiple interacting factors, specific thresholds, instrument details, and the need to synthesize information from different sections of the report. A standard LLM without the text context would likely struggle significantly to answer these questions accurately.